1C4: Analyzing Stars Signals

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AST2000

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October, 2024

Abstract

In this report, we will analyze a set of data from five different stars and use the results to determine whether these stars have planets orbiting them based on their velocities and the change in flux . We also will attempt to determine the mass,radius and the density of these planets.

1 Introduction

Since the dawn of time , we gazed at stars wondering if we were alone in this vast universe and dreaming ,someday, if we will be able to explore other worlds. In the recent few decades, we started facing challenges on our current home planet Earth and potential threat they hold for the survival of our species. In order to survive,we need to be a multi-planetary civilization.That made the searching for habitable planets is a top priority we focused upon the last few decades and harnessed a lot of resources in attempt to achieve that.

In this report, we will use data collected from five stars to determine if they have planets orbiting them. This analyze is based on studying the change in velocity and the flux of a star. When a planet passes in front of a star, it will cause a change in flux coming from this star. Moreover, the presence of a planet orbiting a star, can exert a force on the star causing this star to wobble and rotate around the center of mass .We will use the Doppler effect on the wavelength of the light coming from this star as it rotates around center of mass to evaluate its velocity. Thanks for the advancement of our observational devices , we are able to detect this tiny change in a high precision.

This analyze will allow us to determine the radius , the minimum mass and the density of the planets.

2 Theory

When a system of star and a planet orbiting it and the planet has enough mass to influence the star, the star begins to wobble and rotate around the center of mass. We will assume that the orbit of the star around the center of mass is a perfect circle. As the star rotates around the center of mass , it passes in two phases : toward us and away from us .

When the star is moving toward us, the wavelength of a light will be shorter (Blue shifted). When it moves away from us , the wavelength of a light will be longer (Red shifted) as shown in figure 1

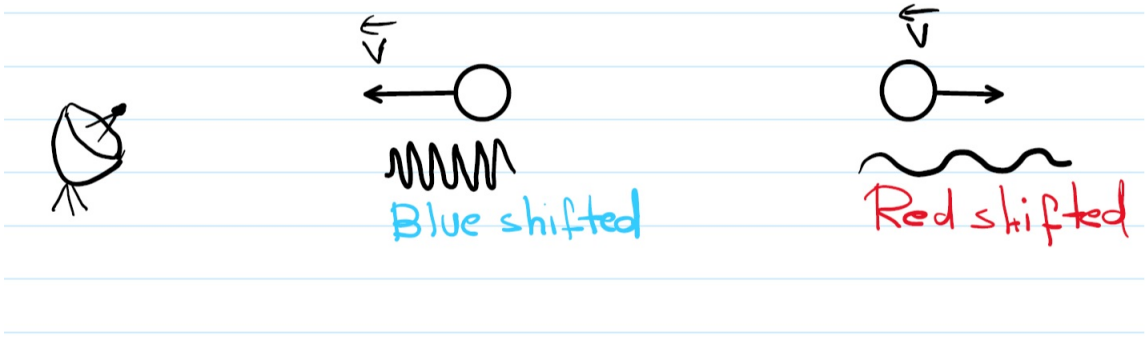


Figure 1: A sketch shows the Doppler effect on a wavelength of observed light from a moving object according to its velocity

This change in the wavelength due to the Doppler effect is given by:

$$\frac{\lambda - \lambda_0}{\lambda_0} = \frac{v_r}{c} \tag{1}$$

Where :

- λ_0: is wavelength as seen from a rest frame of reference.
- λ: is the observed wavelength.
- v_r: the radial velocity
- c: the speed of light

Using this approach, allows us to measure the radial component of the velocity v_r, because only this component contributes to the Doppler effect .

Usually, the orientation of the orbital plane of the star-planet system differs . We will focus only on the plane where has an orbit inclination i which is defined as the angle between the normal vector of the orbital plane with the line of sight for an observer as shown in figure 2 When the inclination i is zero, we won't be able to observe v_r and hence no Doppler effect . As the inclination i increases , the accuracy of measuring v_r improves . When the i approaches 90 degrees , the radial velocity will be equal to the real velocity. the sign of the observed velocity plays an important role of determining whether the system is moving toward or away from us : the positive sign means the distance is increasing and negative means the opposite .

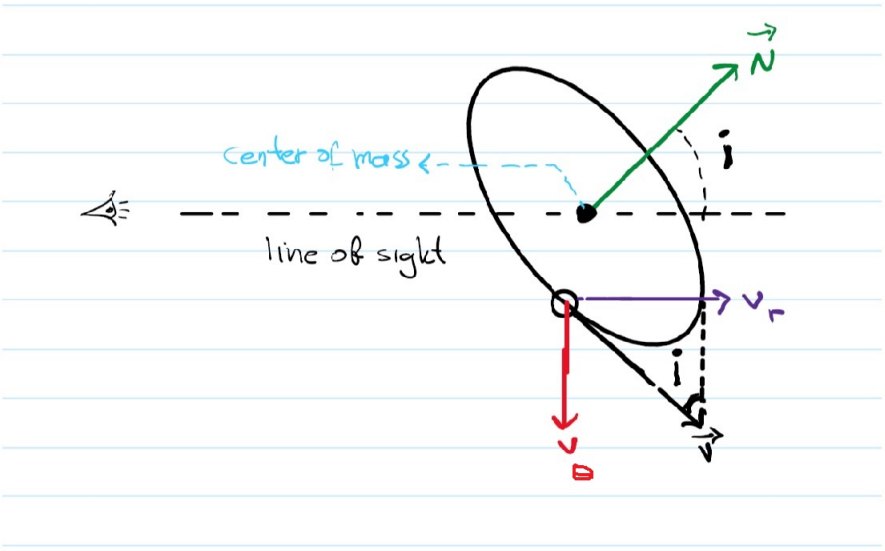


Figure 2: A sketch shows the inclination of an orbital plane of a star with the components of star's velocity

3 Method

The Radial Velocity

From Figure 2 above, we can find using some simple geometry that the radial component is:

$$|v_r| = v \sin i \tag{2}$$

Where :

- v_r: the radial velocity, which is the component of the velocity v taken along the line of sight .
- v: is the velocity of a star around the center of mass.
- i: is the angle between the line of sight and the normal to the orbital plane.

The center of mass of a star-planet system has a constant velocity relative to an observer, we call it "Peculiar velocity"and denoted by v_{pec}. We need to subtract this velocity in order to get the velocity with respect to the center of mass.
Using the position of the star with respect to the observer denoted by $\vec{r}$ is the sum of the position of the center of mass with respect to the observer $\vec{R}$ and the position of the star with respect to the center of mass $\vec{r}_*$:

$$\vec{r} = \vec{R} + \vec{r}_*$$

By taking the derivative of each vector :

$$\vec{r}' = \vec{R}' + \vec{r}'_*$$

we get :

$$\vec{v} = \vec{v}_{pec} + \vec{v}_*$$

The observed velocity$\vec{v}$ is the sum of the peculiar velocity $\vec{v}_{pec}$ and the velocity of the star relative to the center of mass, we call it:$\vec{v}_*$: As illustrated in figure 3

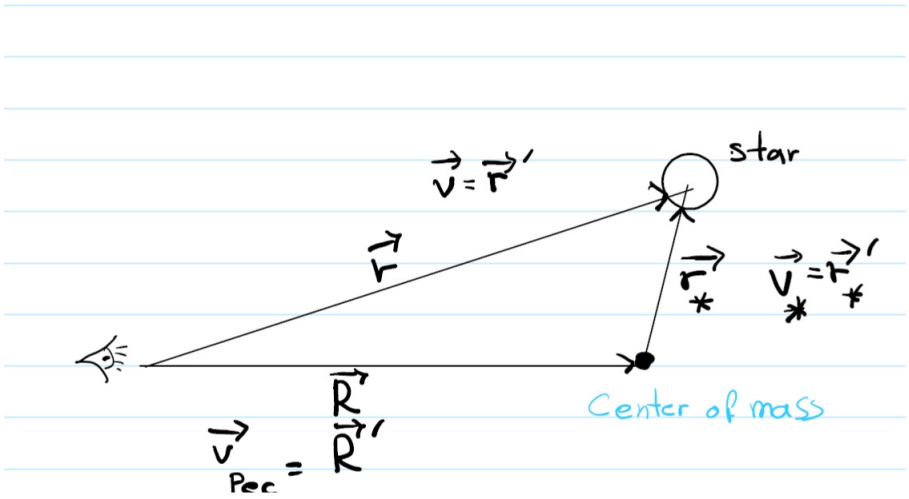


Figure 3: A sketch shows the relation between v ,$\vec{v}_*$ and $\vec{v}_{pec}$

When the star orbit the center of mass, it passes in 4 positions : A,B,C,D as shown in the figure4

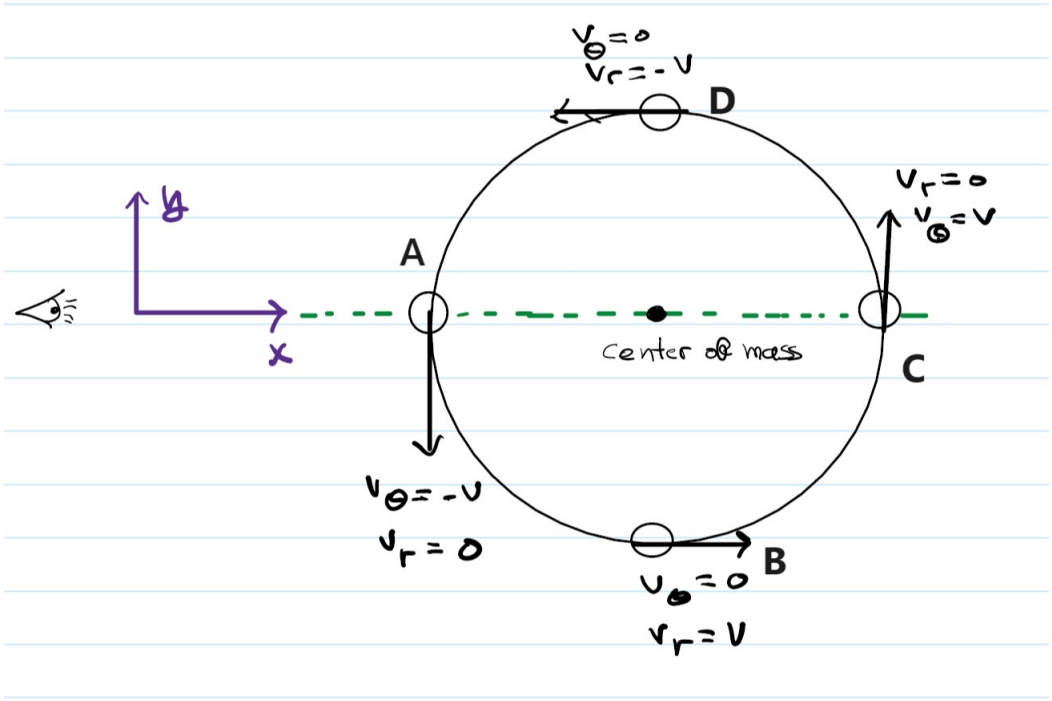


Figure 4: A sketch shows velocity of a star orbiting the center of mass at four positions : A,B,C,D

- At position A, the radial velocity $v_r = 0$ and the observed velocity $\vec{v}$ is equal to $-\vec{v}_\theta$
- At position B, the radial velocity is equal to the observed velocity $v_r = \vec{v}$
- At position C, the radial velocity $v_r = 0$ and the observed velocity $\vec{v}$ is equal to $\vec{v}_\theta$
- At position D, the radial velocity is equal to the observed velocity $v_r = -\vec{v}$

for inclination $i = 90°$ at positions B and D , the radial velocity $v_r$is equal to the real velocity $\vec{v}$. For other inclinations , v_r is given by equation (2)

The Mass Of The Planet

To find the planet mass , we need to combine Kepler third in Newton version : $P^2 = \frac{4\pi a^3}{m_1+M_2}$ with an expression for the a in terms of the velocity of the star and velocity of the planet. In the lectures notes 1C (1), we deduced an formula that finds the mass of the planet . The formula is given :

$$m_p \sin i = \frac{m_*^{\frac{2}{3}} v_{*r} P^{\frac{1}{3}}}{(2\pi G)^{\frac{1}{3}}} \tag{3}$$

Where :

- m_p: is the planet mass.
- m_* : is the star mass.
- v_{*r} is the maximum radial velocity of the star(correspond to point B in figure 4).
- i:the inclination angle.
- P : the period of the star.

The radial velocity as shown in equation (2) is proportional to i: as i approaches 90°,the radial velocity v_{*r} will be greater. Since we can not determine the inclination i, the equation (5) helps us to make a guess about the minimum potential mass of the planet by substituting $i = 90°$ we get the minimum mass of the planetm_p^{min} . If $i < 90°$, the mass Will be greater . The period P and the radial velocity v_{*r}, can be obtained from the plots of the observed velocity over time .

We can observe from the radial velocity curve shown in figure 5, that the curve follows a cosine function.

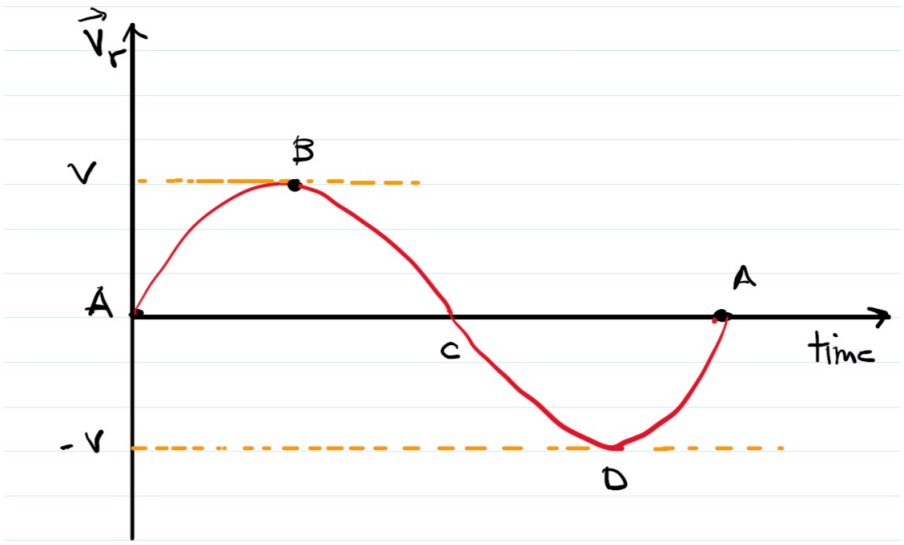


Figure 5: A sketch shows velocity curve of a star orbiting the center of mass at four positions : A,B,C,D

since we assumed the orbit to be a perfect circle, we can write an analytical expression for the radial velocity of the star as :

$$v_{*r}(t) = v_{*r}^B \cos\left(\frac{2\pi}{P}(t - t_0)\right) \tag{4}$$

where :

- v_{*r}: is the radial velocity of the star.
- v_{*r}^B: is the radial velocity at point B.

- P : the period of the star.
- t_0: is defined to be the time when the function has a maximum value($v_{*r}(t) = v_{*r}^B$)

Measuring the radius and the density of the planet

When a planet passes in front of the host star(more precisely , between us and the star.),the planet obscures a small part of the light we receive from the star . When we look at light curve from a star with a planet orbiting it, we can observe a fall in the light intensity as the planet starts entering the disc of the star.When the the planet starts leaving the star disc , the light intensity will start rising again as shown infigure 6 . When we observe a light curve like this , we can predict that the inclination i is approximately 90°.

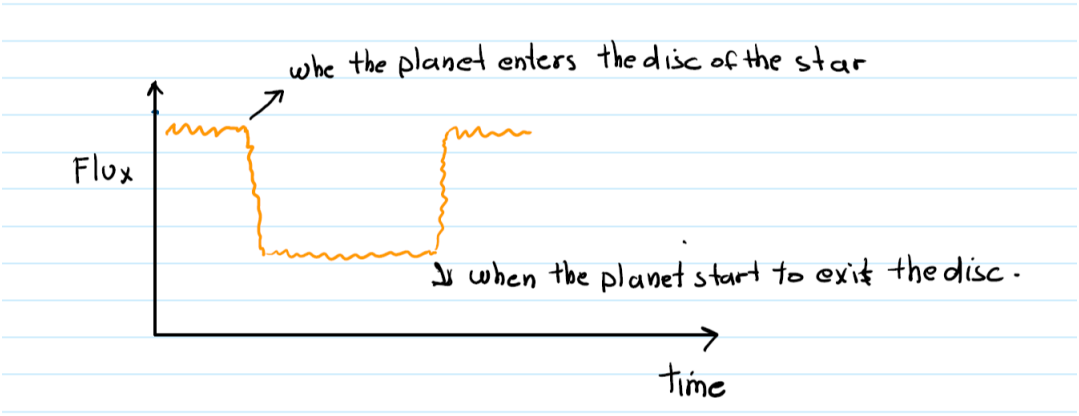


Figure 6: A sketch shows how the flux changes as the planets enters the star disc.

If we are able to detect the time when the planet starts entering the star disc t_0 and the time when the planet is fully entered the star's disc t, we can then measure the radius of the planet. If the radius of the planet is R. When the planet fully entered the star's disc, the planet has passed a distance equal to $2R$ in a time equal to $t - t_0$. The planet moves with a velocity with respect to the star equal to its own velocity v_p with respect to center of mass added to the velocity of the star with respect to center of mass v_*.

By applying the simplest known formula in Physics : distance = speed × time, we get :

$$2R_p = (v_p + v_*)(t - t_0) \tag{5}$$

By using the formula :

$$v_p = v_* \frac{m_*}{m_p}$$

where :m_*,m_p are the mass of the star and the planet respectively. We have that the density is the mass per volume . $V_p = \frac{4}{3}\pi R_p^3$, which gives us :

$$\rho_p = \frac{3m_p}{4\pi R_p^3} \tag{6}$$

Where :

- ρ_p : is the planet density.
- m_p : is the planet mass
- R_p : the planet radius.

The Fit Model

The last thing we need to do is to find a way to determine v_*,P and t_0 precisely for a set of signals with noise. For this purpose , we will use the **Least Square Method** . This method allow us to find the most the most fit curve for the data by minimizing the difference between the data and our model we found in equation (4) and the data.

This model is given by the equation :

$$\Delta(t_0, P, v_*) = \sum_t [v_*^{data}(t) - v_*^{model}(t, t_0, P, v_*)]^2 \tag{7}$$

Where :

- v_*^{data} : is the observed velocity from the data files.
- v_*^{model}: is theoretical model we found in equation (4).

The main idea is to calculate the values of the parameters v_*,P and t_0 that minimize Δ.
When we plot the v_*^{data} from the data , the curves are noisy . The real velocity will be the cosine curve that passes through the middle of this curve as shown in figure 8.
The variables : v_*,P and t_0 can be determined by looking at the velocity curve and make an approximation for the minimum and the maximum values where each parameter possible value is expected to be as shown in the figure 7. These values will be obtained from the curves by eye and inserted into the least square method to obtain the values that minimize the function Δ

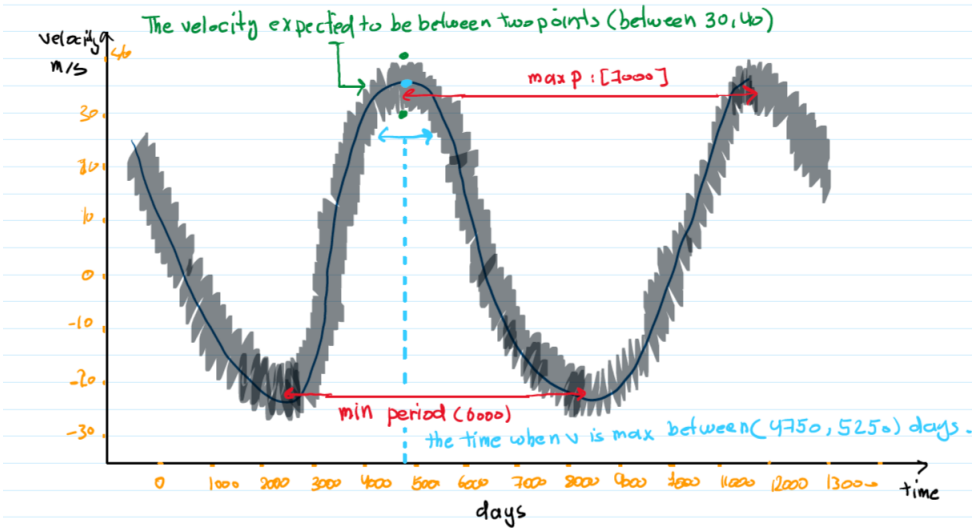


Figure 7: A sketch shows how to determine the max and min values for each parameter where the possible value might be. This is a random data.

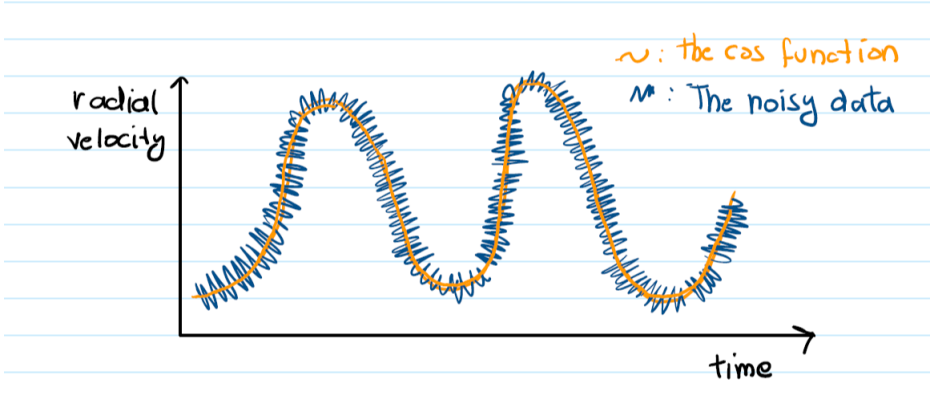


Figure 8: A sketch shows how the idea of the fit model

4 Results

We will analyze the data of the stars one by one:

Star 0

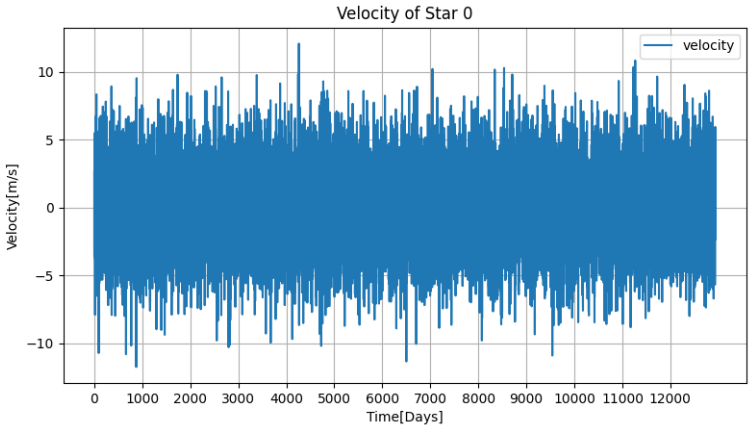


Figure 9: A plot of the velocity of star 0 as function of time

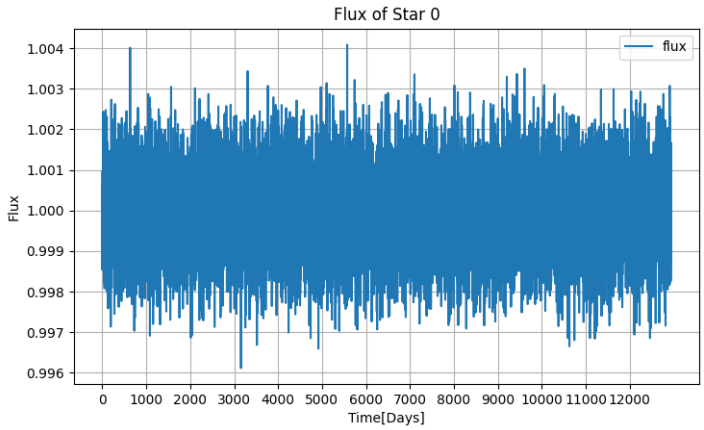


Figure 10: A plot of the flux of star 0 as function of time

By looking at the figure **??** and figure 10, no change neither in the velocity nor in the flux.we can conclude that star 0 has no planets orbiting it.

Star 1

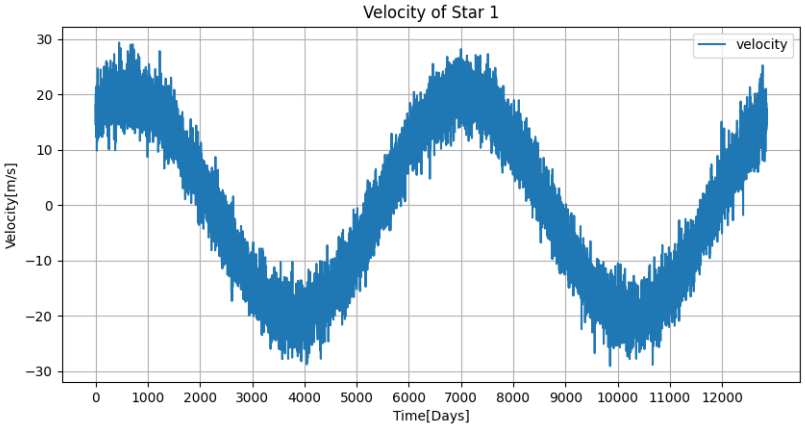


Figure 11: A plot of the velocity of star 1 as function of time

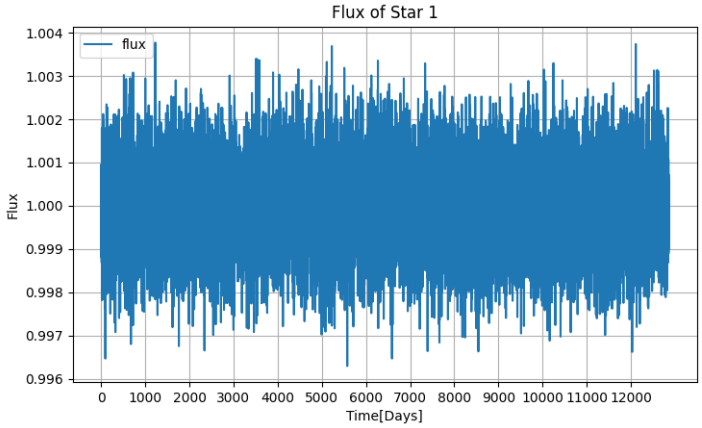


Figure 12: A plot of the flux of star 1 as function of time

By looking at the figure 11 and figure 12, we can observe that there is a change in the velocity and no change in the flux . checking by eye,we could see that the period potentially could be in the interval$(6000, 7000)$ days. The maximum velocity is expected to locate between $(15, 25)$m/s in the time interval $(500, 1500)$ days.

Using the least square model, we got :$P = 6421.05$ days for the period ,$v_* = 20.26$ m/s for the max velocity and $t_0 = (605.26$days for time that the maximum velocity occurs.
To find the minimum mass of planet: we take $\frac{v_*^{min}+v_*^{max}}{2} = \frac{15+25}{2} = 20$ m/s. $P = \frac{6000+7000}{2} = 6500$days. By substituting in equation 3, we get:

$m_p = \frac{1.10^{2/3}\times 20\times 6500^{1/3}}{(2\pi G)^{1/3}} = 3.71 \times 10^{27}$ Kg . For the values from the least square model of $v_* = 20.26$ m/s and$P = 6421.05$ days, we get : $m_p^{model} = 3.74 \times 10^{27}$Kg. By plotting the velocity fitting model with the data , we get the results as in figure 13

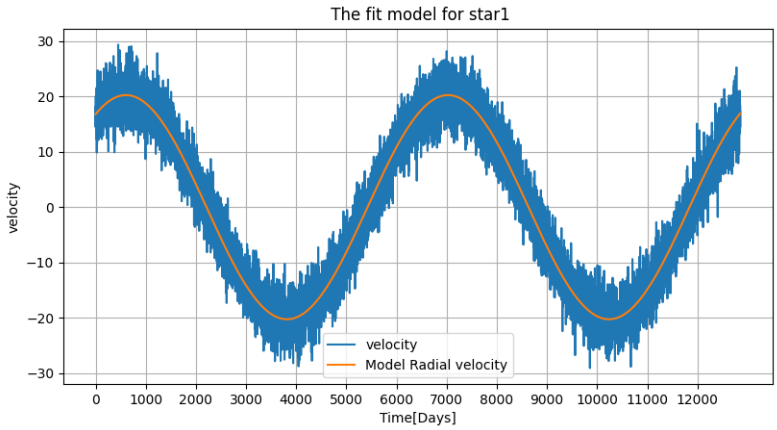


Figure 13: A plot of the velocity of star 1 as function of time with the theoretical model

Star 2

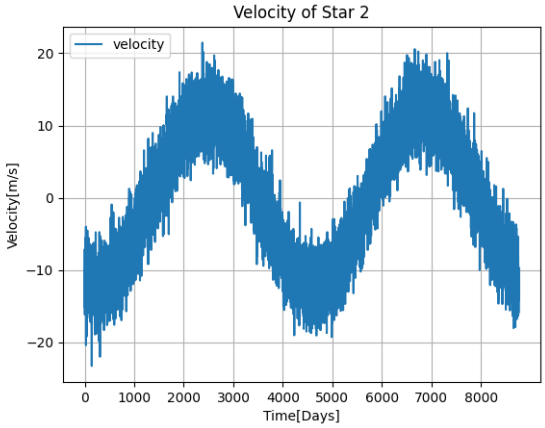


Figure 14: A plot of the velocity of star 2 as function of time

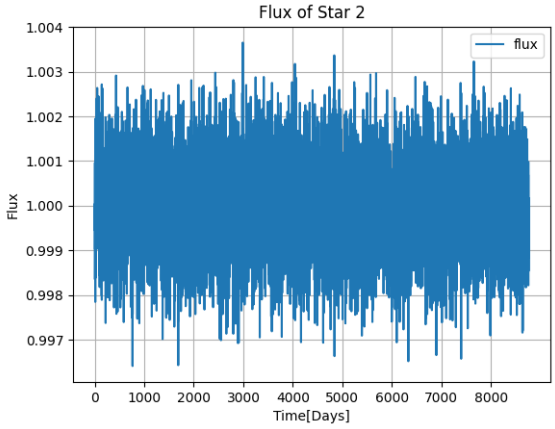


Figure 15: A plot of the flux of star 2 as function of time

We can observe from figure14 and figure 15 a variation in velocity while there is no variation in the flux.
checking by eye,we could see that the period potentially could be in the interval$(4500, 5500)$ days. The maximum velocity is expected to locate between $(5, 15)$m/s in the time interval $(2000, 3000)$ days.

Using the least square model, we got :$P = 4500$ days ,$v_* = 12.36$ m/s for the max velocity and ($t_0 = 2368.42$ days for time that the maximum velocity occurs.
To find the minimum mass of planet: we take $\frac{v_*^{min}+v_*^{max}}{2} = \frac{5+15}{2} = 10$ m/s. $P = \frac{4500+5500}{2} = 5000$days. By substituting in equation (3), we get: $m_p = \frac{0.88^{2/3}\times 10\times 5000^{1/3}}{(2\pi G)^{1/3}} = 1.46 \times 10^{27}$ Kg . For the values from the least square model of $v_* = 12.36$ m/s and$P = 4500$ days, we get : $m_p^{model} = 1.75 \times 10^{27}$Kg.

By plotting the velocity model with the data , we get the results as in figure 16

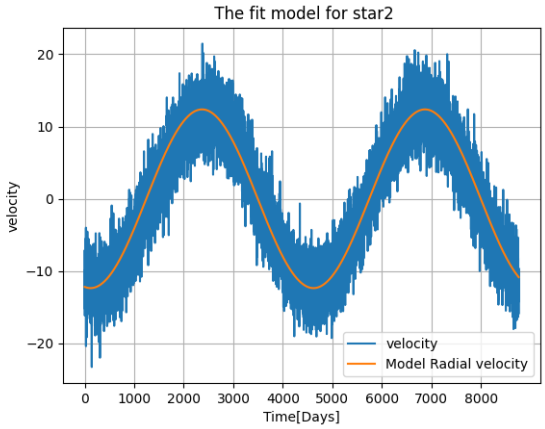


Figure 16: A plot of the velocity of star 2 as function of time with the theoretical model

Star 3

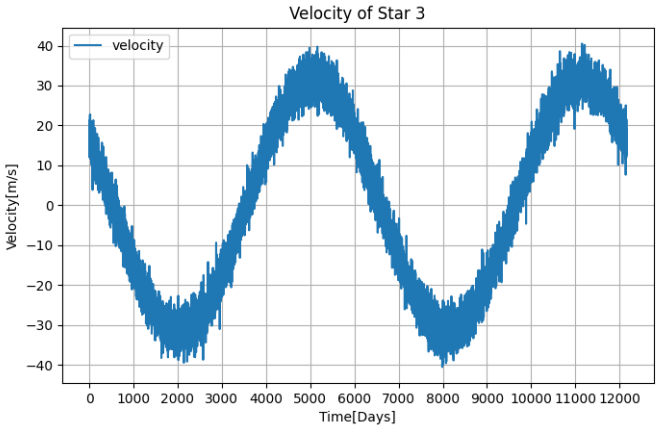


Figure 17: A plot of the velocity of star 3 as function of time

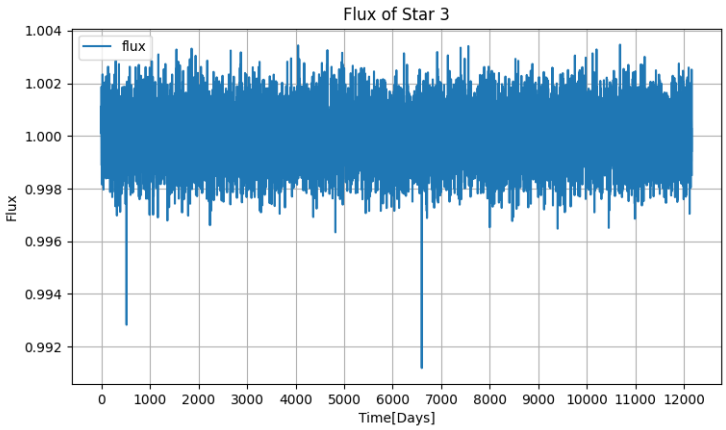


Figure 18: A plot of the flux of star 3 as function of time

We can observe from figure17 and figure 18 a variation in both velocity and in the flux.

checking by eye,we could see that the period potentially could be in the interval$(6000, 6500)$ days. The maximum velocity is expected to located between $(25, 35)$m/s in the time interval $(4500, 5500)$ days. Using the least square model, we got :$P = 6078.94$ days ,$v_* = 31.84$ m/s for the max velocity and $t_0 = 5078.94$days for time that the maximum velocity occurs.

Calculating the period $P = (6000 + 6500)/2 = 6250$days. $v_* = (25 + 35)/2 = 30$m/s.

Inserting them in equation (3), we get : $m_p = 4.84 \times 10^{27}$Kg. For the values from the least square model of $v_* = 31.84$ m/s and$P = 6078.94$ days, we get : $m_p^{model} = 5.09 \times 10^{27}$Kg.

By using the velocity fit model and plot it vs the data for star 3,we get the results : shown in fugue19

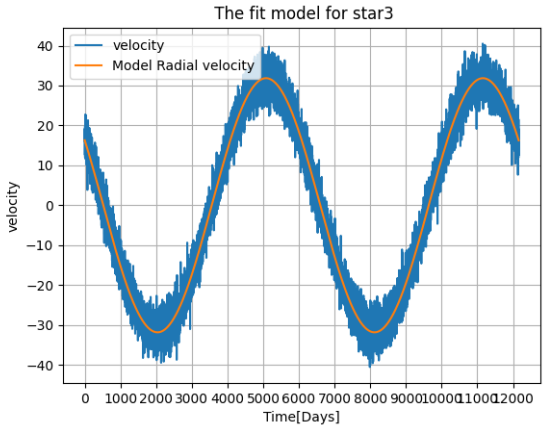


Figure 19: A plot of the velocity of star 3 as function of time with the theoretical model

By looking at the flux plot in figure 18 we try to zoom in in the time interval between (6250,6750) days of the days when the flux starts to fall and plot this interval ,we get results shown in figure :20

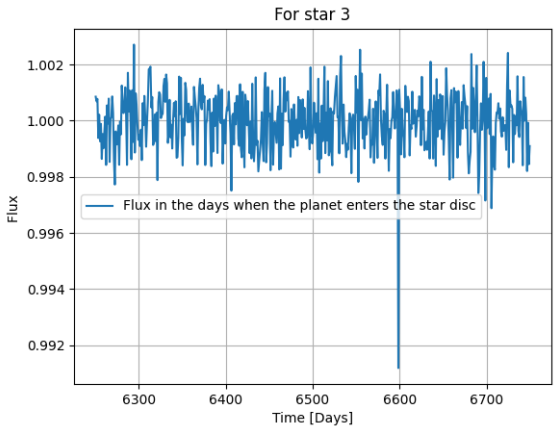


Figure 20: A plot shows the days when the flux for star 3 starts to fall in the time zone 6250,6750 days

Again , zooming in for another time zone (250,750) days in the same plot 18 , we get the figure : 21

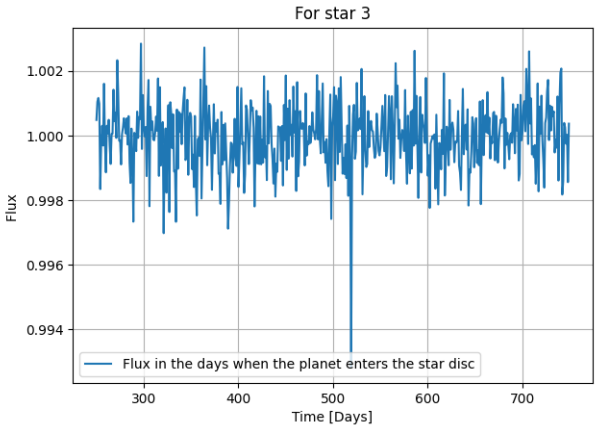


Figure 21: A plot shows the days when the flux for star 3 starts to fall at the zone 250,750 days

we can observe a drop i the flux on day 520 in figure 21 and on day 6600 in figure 20

Star 4

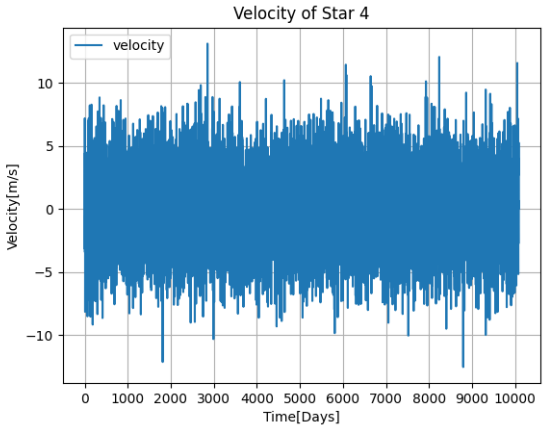


Figure 22: A plot of the velocity of star 4 as function of time

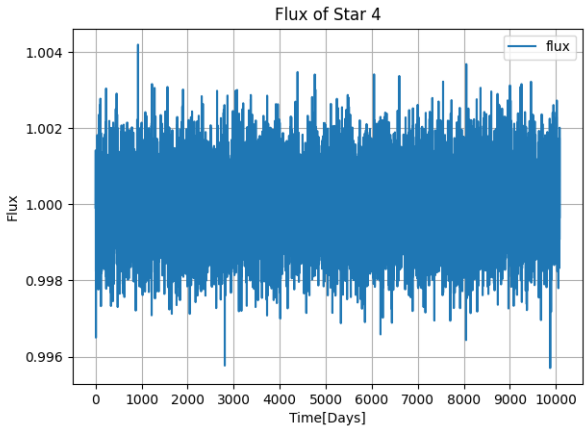


Figure 23: A plot of the flux of star 4 as function of time

By observing 22 and 23, we can see that there is no variation in velocity neither in flux . Results are shown in the table:

Star	Parameter	Value	Unit
Star 1	Mass	1.10	$M_\odot$
	Eye Check Period	6500	days
	Period from Model	6421.05	days
	Eye Check Velocity	20	m/s
	Radial Velocity from Model	20.26	m/s
	Eye Check t_0	(500,1500)	days
	t_0 from Model	605.2	days
	Theoretical Planet Mass	3.71×10^{27}	kg
	Planet Mass from Model	3.74×10^{27}	kg
Star 2	Mass	0.88	$M_\odot$
	Eye Check Period	5000	days
	Period from Model	4500	days
	Eye Check Velocity	10	m/s
	Radial Velocity from Model	12.3	m/s
	Eye Check t_0	(2000,3000)	days
	t_0 from Model	2368.4	days
	Theoretical Planet Mass	1.46×10^{27}	kg
	Planet Mass from Model	1.75×10^{27}	kg
Star 3	Mass	0.91	$M_\odot$
	Eye Check Period	6250	days
	Period from Model	6078.9	days
	Eye Check Velocity	30	m/s
	Radial Velocity from Model	31.8	m/s
	Eye Check t_0	(4500,5500)	days
	t_0 from Model	5078.9	days
	Theoretical Planet Mass	3.84×10^{27}	kg
	Planet Mass from Model	5.09×10^{27}	kg

Table 1: Parameters for Star 1, Star 2, and Star 3

5 Discussion

From our results, it is clear that stars 1, 2, and 3 have planets orbiting them due to the variation in their velocities. For the stars 0 and 4, we can not decide whether they have planets orbiting them or

not . there may be planets but the inclination of their orbiting plane not aligned with our sight line and that prevents us from observing any radial velocity .

For star 1: There was a close agreement in the period, velocity, and t_0 between the eye check and the results from the least square method. The fitting model plot is almost located in the middle of the thick line which is an indication of the good choice of the upper and lower bounds of the parameters. The numerical estimation of the planet mass gave us a result $m_p^{model} = 3.74 \times 10^{27}$Kg and eye-check result $m_p = 3.71 \times 10^{27}$Kg showing high accuracy .

For star 2 : There was a close agreement in the period, velocity, and t_0 between the eye check and the results from the least square method. The fitting model plot is almost centered in the middle of the thick line. . In the estimation of the planet mass, the numerical method resulted $m_p^{model} = 1.75 \times 10^{27}$Kg. and check by eye result $m_p = 1.46 \times 10^{27}$Kg, gave a small deviation with a ratio 1.19 . This could be related to the deviation between eye check values for velocity and period from the values obtained by the model .

For star 3 : There was a close agreement in the period, and t_0 between the eye check and the results from the least square method. The resulted mass of the planet was with a ratio of 1.05 between the numerical and eye-check method. By analyzing the zoomed flux spectrum for the flux plot of star3 in 21 and 20, we see the drop in the flux occurs in two points day$\approx$ 6600 and day $\approx$ 520 respectively .
This result agrees with the theoretical framework which states that the radial velocity should be zero every half period,specifically at points A and C in figure 4 . However, we observe the fall in flux every full period in this plot .The reason is we can not observe the change in flux when the planet passes in front of the star on the other side . If we look at figure 18, we see that the two falls in flux are spaced by a time difference that is indeed equal to one full period 6078.94 for star 3.
TThe method used in the estimation of the radial velocity is the **Least Square Method** . In this method we had to analyze a curve an chose upper and lower limit for the velocity where it is most possible value might be. The least squares method minimizes the sum of the squares of the difference between the observed values and the model predictions. This method is useful because it provides the most statistically likely parameter estimates for the data, especially when dealing with observational with noisy data.. The use of number of steps N= 20 might not be the optimal choice but it still a good choice . reducing it to less than 20 , it will affect the accuracy in a remarkable way while increasing it could improve the accuracy but it will significantly increase computational time and resource usage ad exhaust the computer. To estimate the mass of the planets we used also similar method where we have to set a lower bound for the mass .The alignment between the results from eye-watch method and the least square is acceptable .

6 Conclusion

We concluded that stars 1,2,3 have planets orbiting them. The estimation of planets mass for 1,3, gave us that the calculated masses were very close to the masses obtained by the model . From this we may conclude that the inclination angle of these stars are both likely to be 90°. For star 2,the difference between calculated and the mode mass means that the inclination of this this orbital plane is less than 90°. In summery , analyzing the plots with eye is not a precise tool to make conclusion, but it provides a useful starting point.However, combining it with the least square method gave a good approximation of the data and yielded reasonable results. Least Square Method can be useful and a reliable tool to make conclusions in further similar cases.

References

[1] Frode Kristian Hansen. Lecture Notes 1C *Introduction to Astrophysics,AST2000.*